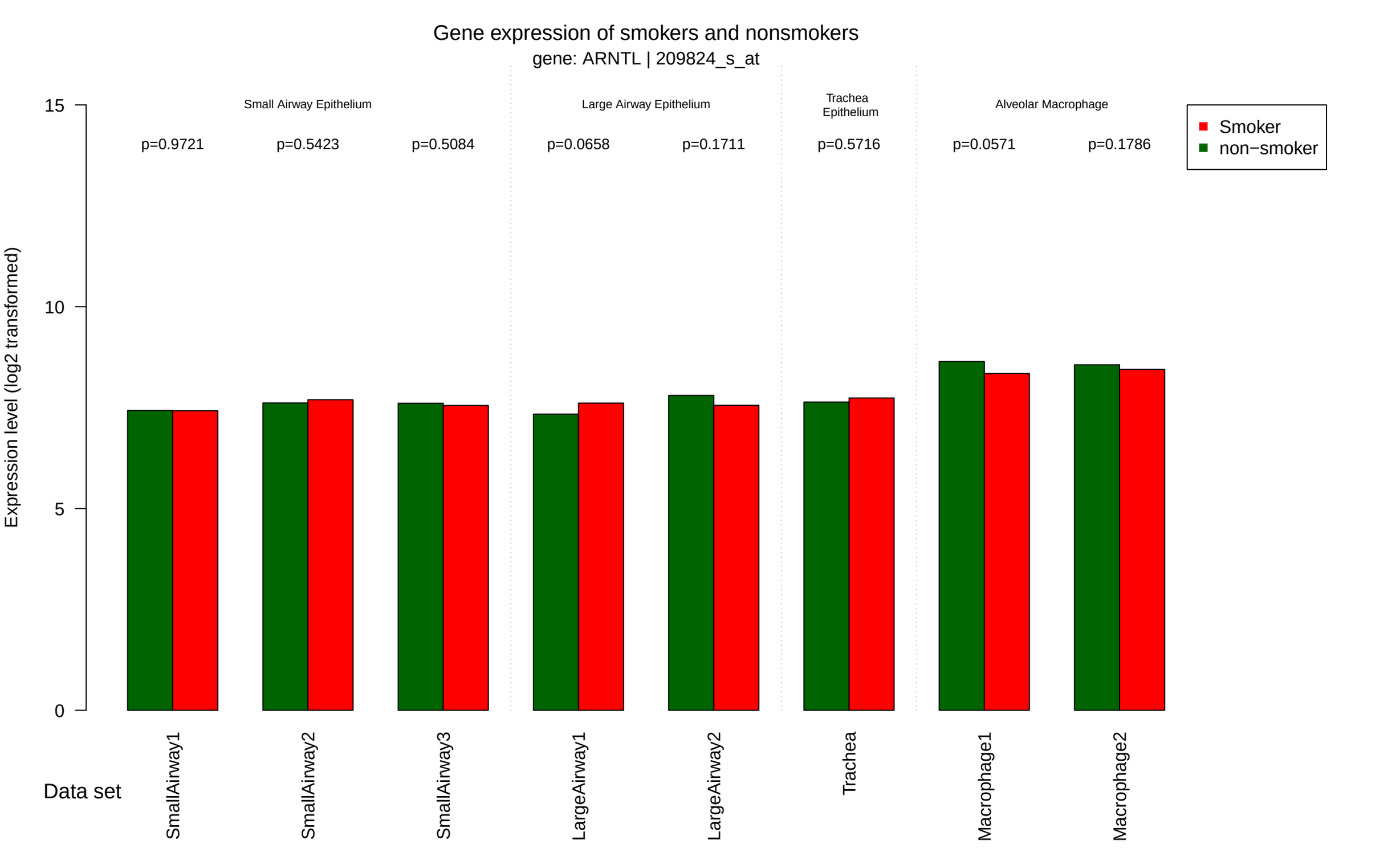
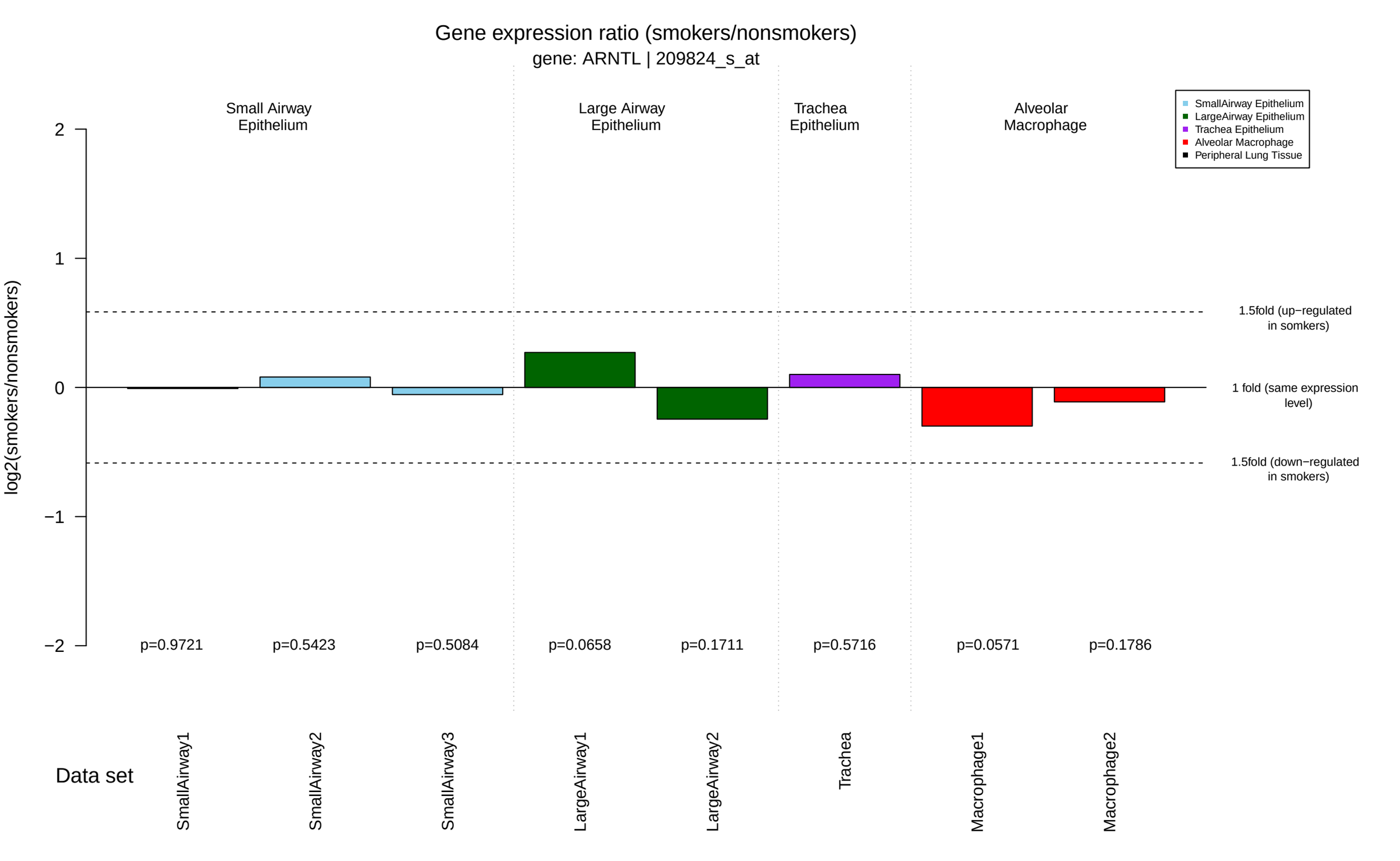
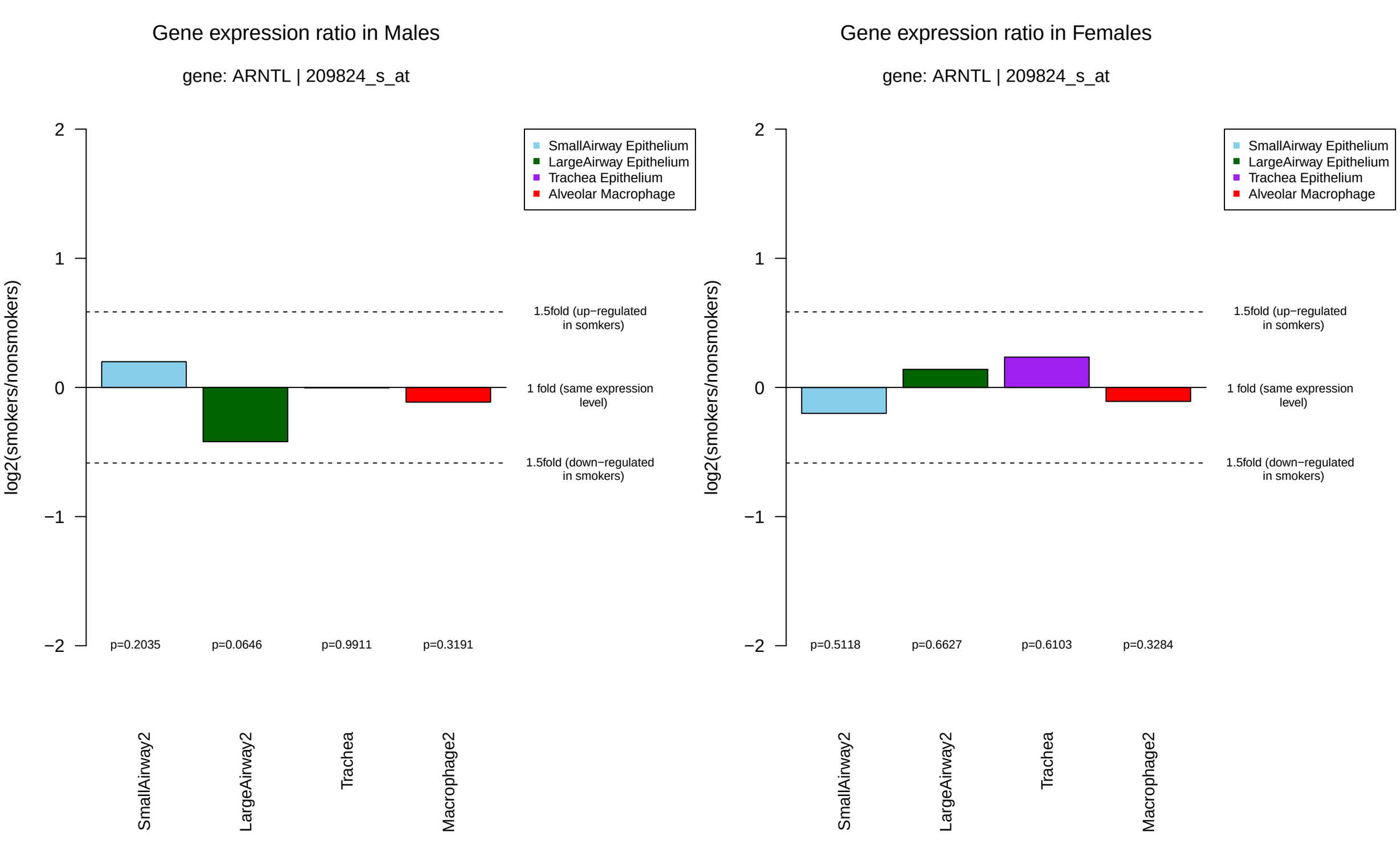
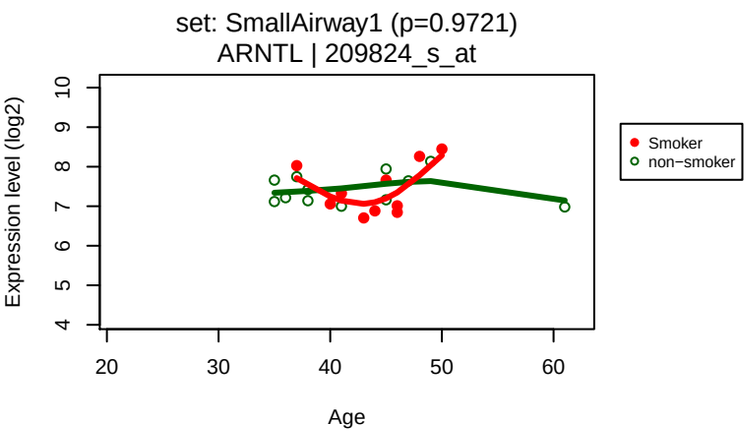
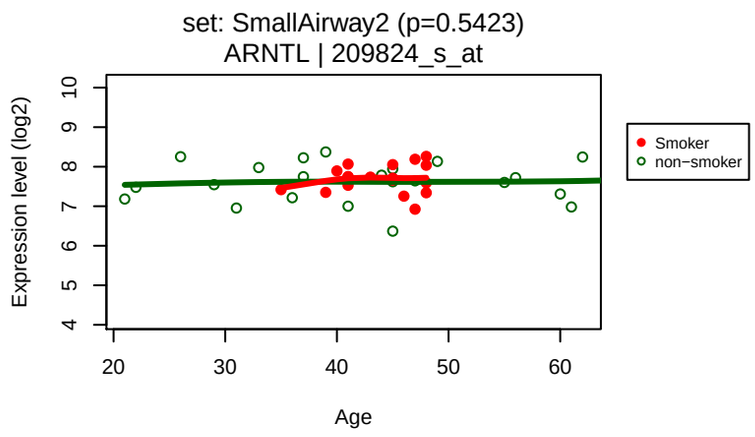
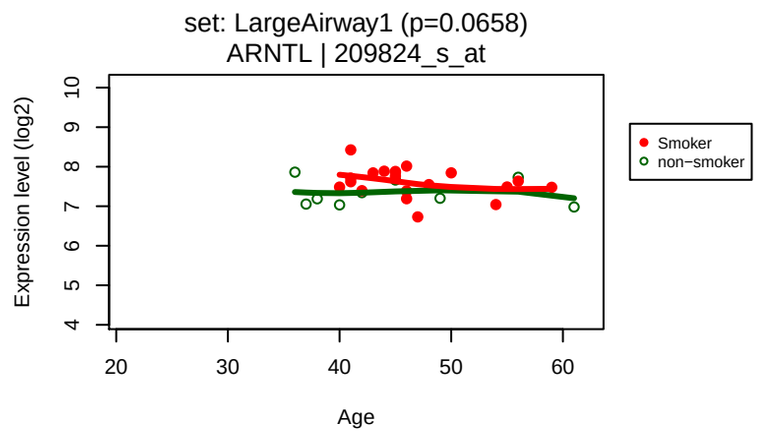

Gene expression of smokers and nonsmokers
gene: ARNTL | 209824_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9721
p=0.5423
p=0.5084
p=0.0658
p=0.1711
p=0.5716
p=0.0571
p=0.1786
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

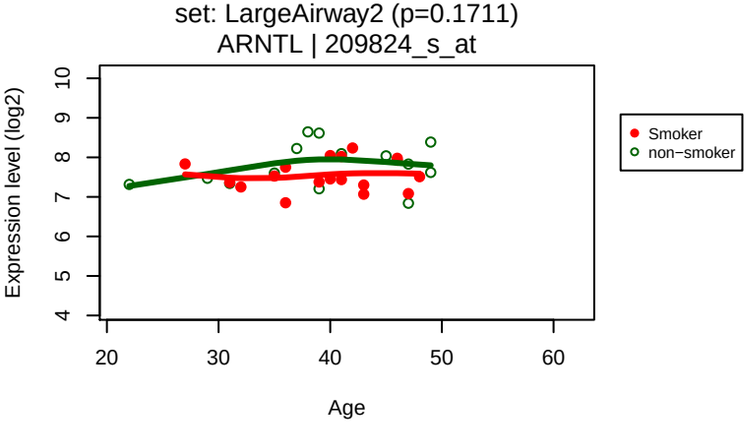

Gene expression ratio (smokers/nonsmokers)
gene: ARNTL | 209824_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9721
p=0.5423
p=0.5084
p=0.0658
p=0.1711
p=0.5716
p=0.0571
p=0.1786
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

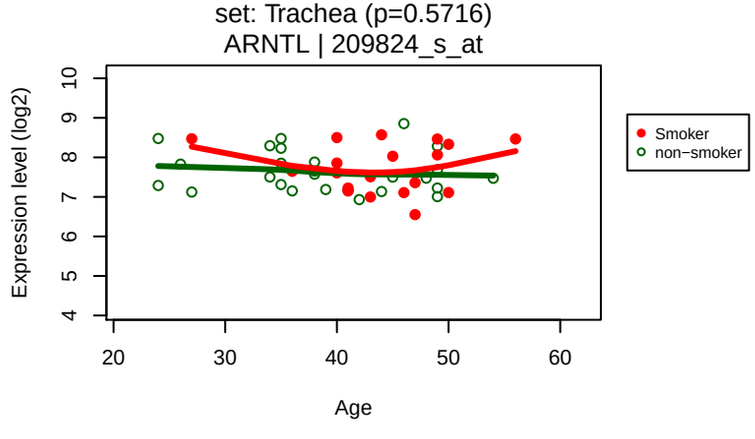

Gene expression ratio in Males
gene: ARNTL | 209824_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2035
p=0.0646
p=0.9911
p=0.3191
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARNTL | 209824_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5118
p=0.6627
p=0.6103
p=0.3284
SmallAirway2
LargeAirway2
Trachea
Macrophage2

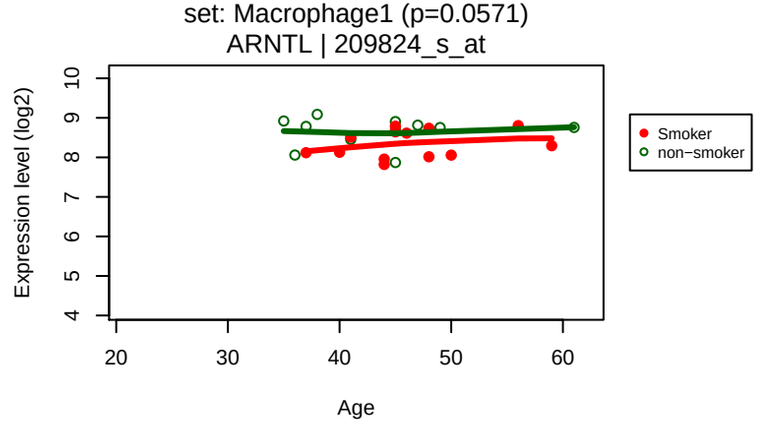

set: SmallAirway1 (p=0.9721)
ARNTL | 209824_s_at
Expression level (log2)
Age
Smoker
non−smoker

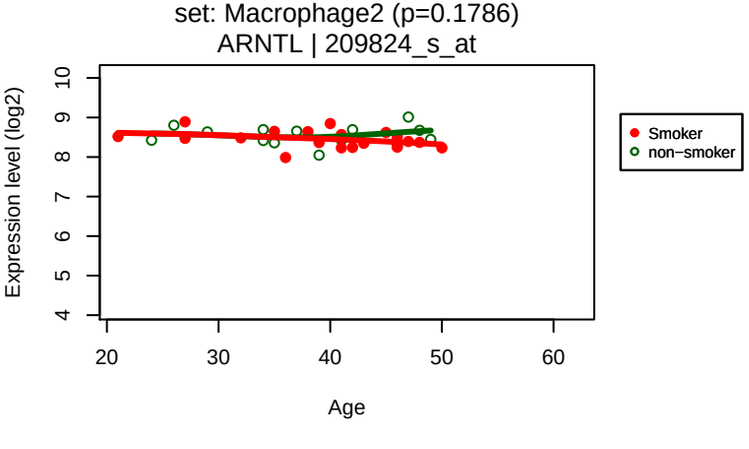

set: SmallAirway2 (p=0.5423)
ARNTL | 209824_s_at
Expression level (log2)
Age
Smoker
non−smoker

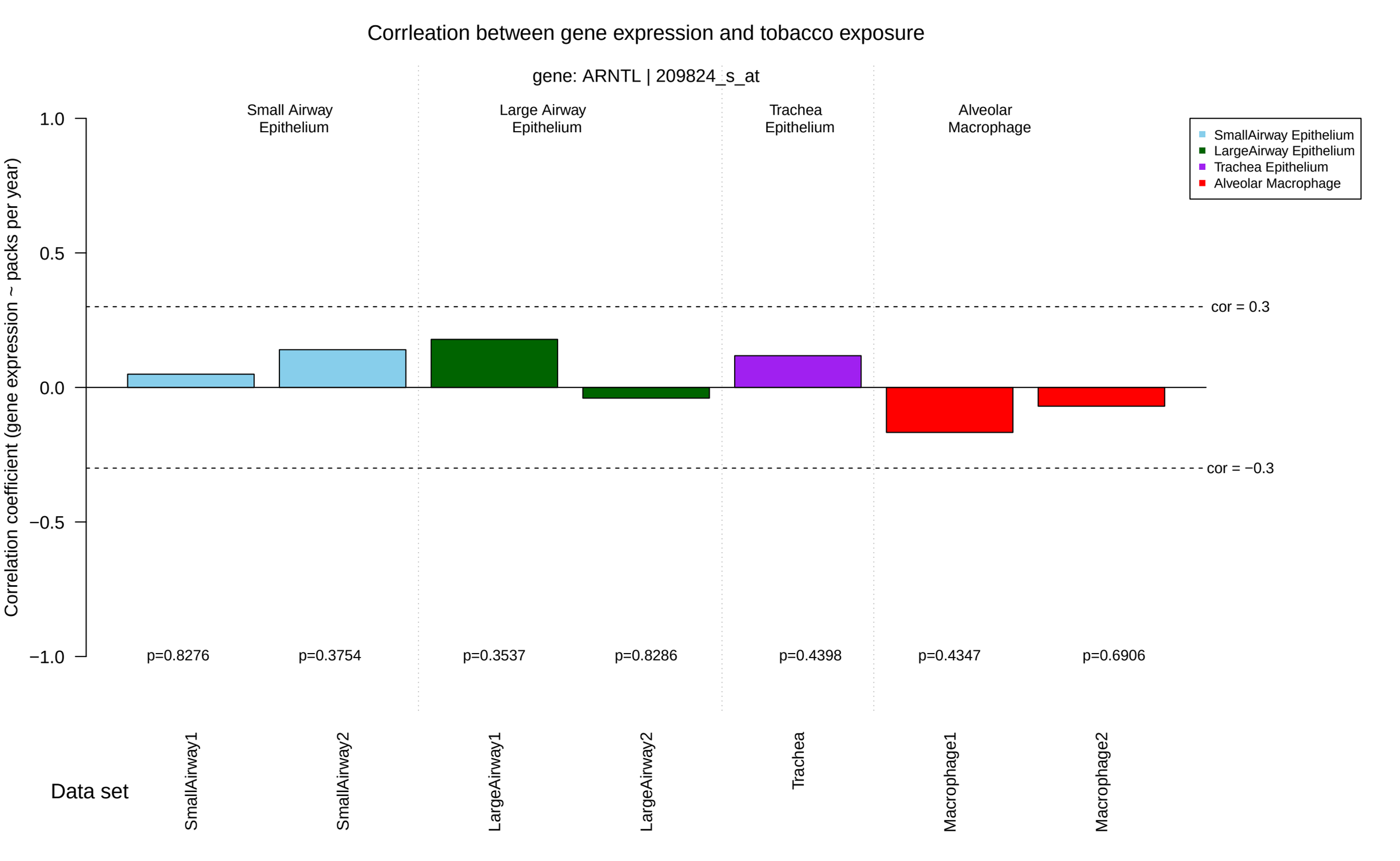

set: LargeAirway1 (p=0.0658)
ARNTL | 209824_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1711)
ARNTL | 209824_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5716)
ARNTL | 209824_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0571)
ARNTL | 209824_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1786)
ARNTL | 209824_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARNTL | 209824_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8276
p=0.3754
p=0.3537
p=0.8286
p=0.4398
p=0.4347
p=0.6906
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2